Tannu Kumari

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PROFESSIONAL SUMMARY

B.Tech Computer Science student (Computer Graphics & Visualization) at Sikkim Manipal Institute of Technology with a strong foundation in software development and machine learning. Proficient in Python, C++, SQL. Completed a research internship at NIT Patna, working on a Smishing Detection project using machine learning. Passionate about leveraging technology to solve real-world problems.

EDUCATION

Year	Degree	Institution	CGPA/Score
2023-2027	**B.Tech, CSE (Graphics & Visualization)**	Sikkim Manipal Institute of Technology	8.9/10
2021-2022	Class XII, CBSE	Gurukul Vidyapeeth, Rampur, Hajipur	90%
2019-2020	Class X, CBSE	St. John's Academy, Hajipur	88%

WORK EXPERIENCE

Research Intern - National Institute of Technology, Patna *May 2025 - July 2025*

- Developed a machine learning system for detecting smishing (SMS phishing) attacks with real-time classification.
- Applied TF-IDF feature extraction and trained models including SVM, Random Forest, and Logistic Regression, achieving up to 99% accuracy.
- Handled class imbalance using oversampling and improved model robustness against obfuscated attacks.
- Evaluated models using F1-score and ROC-AUC for performance optimization.
- **Technologies:** Python, Scikit-learn, Pandas, NumPy

PROJECTS

Language-Aware Hybrid Sentiment Analysis on Twitter Data *August 2024 - April 2026*

- Developed a language-aware hybrid sentiment analysis system to classify Twitter posts into Positive, Negative, and Neutral sentiments across English and multilingual text.
- Implemented a hybrid inference pipeline using LangID for automatic language detection, routing English tweets to BERTweet and multilingual tweets to XLM-RoBERTa, with combined prediction for mixed-language inputs.
- Fine-tuned transformer-based models on English and multilingual sentiment datasets and evaluated performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
- **Technologies Used:** Python, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, NLTK, LangID.

Smishing Detection using Machine Learning *May 2025 – July 2025*

- Built an on-device SMS phishing detection system using TF-IDF and ML models.
- Trained classifiers including SVM and Random Forest, achieving up to **99% accuracy**.
- Addressed class imbalance using oversampling and enhanced robustness against obfuscated attacks.
- Evaluated model performance using ROC-AUC and F1-score.
- **Technologies Used:** Python, Scikit-learn, Pandas, NumPy

ACHIEVEMENTS & CERTIFICATIONS

- **Runner-up – UI/UX Hackathon (Tech Adrista '24):** Conducted by MLSC Club
- **3rd Rank – Smart India Hackathon (SIH) 2024:** Secured third place in the inter-college round, Punjab Government Portal track.
- **Event Manager Lead – Club Artistic, The Five Elements Club SMIT:** Led cultural and technical events.
- **Winner – Rang Tarang (2024 & 2025):** Secured 1st place in Rangoli Competition during SMIT's cultural fest.
- **Workshop – AI Tools Bootcamp**
- **Oracle Certified - AI Foundations Associate – OCI (2025)**

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C/C++, SQL, HTML5, CSS3
- **Machine Learning:** Scikit-learn, Pandas, NumPy, NLTK, Imbalanced-learn, XGBoost, TF-IDF, PyTorch, BERTweet
- **Databases:** MongoDB, MySQL
- **Platforms & Tools:** Git, GitHub, Kaggle
- **Development Environments:** VS Code, Visual Studio